Ziyi Wang

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Professional Summary

MEng Mechanical Engineering student at the University of Toronto and BEng Materials Engineering graduate from McGill University, with industry experience in mechanical design, process optimization, and engineering analysis across semiconductor refining, advanced composites, and steelmaking operations. Proficient in SolidWorks, AutoCAD, ANSYS Fluent, MATLAB, Microsoft Visio, and Microsoft Project. Canadian citizen, and eligible for Engineer-in-Training registration.

Education

Bachelor of Engineering: Materials Engineering – McGill University, Montreal, QC *Sep 2021 – Dec 2025*
Master of Engineering: Mechanical Engineering — University of Toronto, Toronto, ON *Sep 2026 – Sep 2027*

Engineering Experience

Process Engineering Intern | 5N Plus Inc. – Montreal, QC Jun 2025 – Oct 2025

High-Purity Semiconductor Metal Refining – Yield Optimization & Process Contamination Control

- Led contamination root-cause analysis across high-purity (distillation) and ultra-high-purity (zone refining, up to 99.9999%) processes for Cd, Te, Zn, Sb, and CdTe; documented findings in process flow diagrams (PFDs) and updated P&IDs to reflect corrected control points.
- Designed and executed improved product sampling strategies and clean room environmental testing; analyzed ICP OES/MS data to map contamination distribution within product ingots and optimize process parameters and control strategies, **achieving a 40% increase in cadmium purification line yield.**
- Performed techno-economic evaluation of antimony purification routes (multi-stage zone refining vs. external supply); identified an optimized process flow that **increased yield from 26.5% to 42.4% while reducing costs by over 45%.**

Composite Engineering Intern | Linamar Corporation – Guelph, Ontario Sep 2024 – Dec 2024

Resin Additive Integration for Performance Enhancement of Carbon Fiber Hydrogen Storage Tanks

- Developed P-SiO_2-enhanced resin formulations for Type IV carbon fiber hydrogen tanks via structured Design of Experiments (DOE); optimized additive dispersion and curing parameters, **achieving an average burst pressure improvement of over 2,000 psi per tank chamber after additive integration.**
- Collaborated with external synthesis lab on sol–gel production of P-SiO_2 additives; validated resin and tank performance via SEM, FTIR, DSC, TGA, hydrostatic burst, hydraulic cycling, tensile, and interlaminar shear testing.
- Prepared AutoCAD technical drawings for tank production, including production line layouts, polymer liner geometries, fiber winding schematics, laminate layer layouts, and sectional views to support manufacturing communication and process documentation.

R&D Engineering Intern | EVRAZ North America – Regina, Saskatchewan Jan 2023 – Aug 2023

Steelmaking Vacuum Degassing Process Modeling (Published – AIST, 2024, Link: doi.org/10.33313/388/077)

- Developed an effective equilibrium reaction-zone model to simulate and predict nitrogen removal from molten steel in a vacuum tank degasser; designed process flow logic, PFDs, and mass and energy balance calculations from first principles; programmed automated iterative calculations in FactSage across varying pressure and process time.
- Modelled 3D geometries of the vacuum tank degasser (ladle, snorkel, gas injection regions) in SolidWorks and integrated into ANSYS Fluent for Computational Fluid Dynamics (CFD) simulations of multiphase steel–argon flow, analyzing turbulence, bubble plume dynamics, and mass transfer behavior during de-nitrogenation.

Capstone Project - Aerospace Turbine Blade Thermal Coating Design

- Engineered a functionally graded air-plasma-sprayed coating system for GE Aerospace CF6 turbine blades.
- Developed coating architecture (NiCrAlY bond coat + YSZ gradient layers) to mitigate thermal expansion mismatch, enhance strain tolerance, and reduce thermal conductivity under cyclic high-temperature exposure.
- Integrated process flow and mass/energy balances to validate coating throughput and feasibility.

Technical Skills

- **Engineering Software:** SolidWorks | AutoCAD | Fusion 360 | ANSYS Fluent | FactSage | Thermo-Calc | MS Visio | MS Project | MATLAB | Python | Excel VBA
- **Material Testing & Characterization:** ICP-OES/MS | SEM | XRD | FTIR | DSC| Hydrostatic Burst | Hydraulic Cycling | Tensile & Interlaminar Shear | Hardness & Fracture Analysis | Creep Evaluation | Finite Element Analysis (FEA)